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Cryptography Assignment - 3

- 1) Determine the multiplication inverse module 26 for each of the following integers

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

$$3^{-1} \bmod 26$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1) - \textcircled{1}$$

$$2 = 26 + 3(-8) - \textcircled{2}$$

$$1 = 3 + 2(-1)$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(8) + 26(-1) = 1$$

$$3^{-1} = 9$$

$$5^{-1} \bmod 26$$

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$5^{-1} = 21$$

7^{-1}

$$26 = 7(3) + 5$$

$$5 = 26 + 7(-3)$$

$$7 = 5(1) + 2$$

$$2 = 27 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$5 + 2[7 + 5(-1)](-2) = 1$$

$$5 + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} = 15$$

 9^{-1}

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = 3$$

11^{-1}

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + (26 + 11(-2))(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-3) = 1$$

$$26 - 7 = 19$$

$$11^{-1} = 19$$

 15^{-1}

$$26 = 15(1) + 11$$

$$11 = 26 + 15(-1)$$

$$15 = 11(1) + 4$$

$$4 = 15 + 11(-1)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 - 3(-1) = 1$$

$$15 + 11(-1) + (11 + 4(-2))(-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + (26 + 15(-1))(-2) + (15 + 11(-1))(2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15 + (5) + 26(-2) + 26 + 15(-1)(-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$15^{-1} = 7$$

$$17^{-1}$$

$$26 = 17(1) + 9$$

$$9 = 26 + 17(-2)$$

$$17 = 9(1) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$26 + 17(-1) + (17 + 9(-1))(-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$17^{-1} = 23$$

$$19^{-1}$$

$$26 = 19(1) + 7$$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$19 + 7(-2) + (7 + 5(-1))(-2) = 1$$

$$19 + 7(-2) + 7(-2) + 5(2) = 1$$

$$19 + 7(-4) + 5(2) = 1$$

$$19 + (26 + 19(-1))(-4) + (19 + 7(-1))(-2) = 1$$

$$19 + 26(-4) + 19(4) + 19(2) + 7(-4) = 1$$

$$19(7) + 26(-4) + 26(4) + 19(4) = 1$$

$$19(11) + 26(-8) = 1$$

$$19^{-1} = 11$$

$$21^{-1} \quad 26 = 21(1) + 5 \quad 5 = 26 + 21(-1)$$

$$21 = 5(4) + 1 \quad 1 = 21 + 5(-4)$$

$$21 + 5(-4) = 1$$

$$21 + (26 + 21(-1))(-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$26(-4) + 21(5) = 1$$

$$21^{-1} = 5$$

$$25^{-1}$$

$$26 = 23(1) + 3$$

$$23 = 2(7) + 2$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$26 + 23(-1) + (23 + 2(-7))(-1) = 1$$

$$26 + 23(-2) + 26(7) = 1$$

$$26 + 23(-1) + 23(-1) + (26 + 23(-1)) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$25^{-1} = 17$$

$$25^{-1}$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$25^{-1} = 26$$

2) Choose any string of your choice to encrypt using the affine cipher with key value multiplication 5 & addition 8

2) Encryption: AYUSHMAN using Affine Cipher

multiplication $a=5$

Addition $b=8$

$$E(x) = (ax + b) \bmod 26$$

$$E(x) = (5x + 8) \bmod 26$$

So, similarly

Letter	A	Y	U	S	H	M	A	N
n	0	24	20	18	7	12	0	13

Apply $5x+8$

$$A: (5(0) + 8) \bmod 26 = 8 = I$$

$$Y: (5(24) + 8) \bmod 26 = 128 \bmod 26 = 24 = Y$$

$$U: (5(20) + 8) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$S: (5(18) + 8) \bmod 26 = 98 \bmod 26 = 20 = U$$

$$H: (5(7) + 8) \bmod 26 = 43 \bmod 26 = 17 = R$$

$$M: (5(12) + 8) \bmod 26 = 68 \bmod 26 = 16 = Q$$

$$A: 8 = I$$

$$N: (5(13) + 8) \bmod 26 = 73 \bmod 26 = 21 = V$$

AYUSHMAN $\rightarrow$ IYEURQIV

3) Encrypt "HI" using the Affine cipher with $a=11$ & $b=5$

$$a=11, b=5$$

formula:

$$E(x) = (11x + 5) \bmod 26 = 4 = H$$

$$E(H) = (11(7) + 5) \bmod 26 \\ = 82 \bmod 26 = 4 = E$$

$$I = 8$$

$$E(I) = (11(8) + 5) \bmod 26 \\ = 93 \bmod 26 = 15 = P$$

$$HI \Rightarrow EP$$

4) The ciphertext UCR was encrypted using the affine function $9x + 2 \bmod 26$. Find the ~~plaintext~~ plaintext.

$$C = (9x + 2) \bmod 26$$

$$9^{-1} \bmod 26$$

$$26 = 9(2) + 8$$

$$8 = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = 3$$

Decryption formula: $x = 3(C - 2) \bmod 26$

$$U = 20 \quad x = 3(20 - 2) = 54 \bmod 26 = 2 = C$$

$$C = 2 \quad x = 3(2 - 2) = 0 = A$$

$$R = 17 \quad x = 3(17 - 2) = 45 \bmod 26 = 19 = T$$

$$19 = T$$

UCR $\Rightarrow$ CAT

5) Decrypt the cipher text using Affine cipher, K D Y X J U W D E X Z Y using the Key 3 (Multiplm) and 4

5) $a=3, b=4$

$$P = a^{-1}(c-b) \bmod 26$$

K D Y X J U W D E X Z Y

10 3 24 23 9 20 22 3 4 23 25 24

$$a^{-1} = 3^{-1}$$

$$26 = 3(8) + 2 \quad 2 = 26 + 3(-8)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + (26 + 3(-8))(-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$3^{-1} = 9 = a^{-1}$$

$$K = 9(10-4) \bmod 26 = 9(6) \bmod 26 = 2 = C$$

$$D = 9(3-4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$Y = 9(24-4) \bmod 26 = 9(20) \bmod 26 = 24 = Y$$

$$X = 9(23-4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$J = 9(9-4) \bmod 26 = 9(5) \bmod 26 = 19 = T$$

$$U = 9(20-4) \bmod 26 = 9(16) \bmod 26 = 14 = O$$

$$W = 9(22-4) \bmod 26 = 9(18) \bmod 26 = 6 = G$$

$$D = 9(3-4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$E = 9(4-4) \bmod 26 = 9(0) \bmod 26 = 0 = A$$

$$X = 9(23-4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$Z = 9(25-4) \bmod 26 = 9(21) \bmod 26 = 7 = H$$

$$Y = 9(24-4) \bmod 26 = 9(20) \bmod 26 = 24 = Y$$

Plaintext: CRYPTOGRAPHY

- 6 Use $K_1 = 13$ (multiplier), $K_2 = 4$ (Even though it is not a valid key - try to encrypt). Encrypt the string "input" and also using the above key and find the what is the issue.

$$C = (P \times \text{Multi} + \text{Add}) \bmod 26$$

$$C = (P \times a + b) \bmod 26$$

$$a = 13 \quad b = 4$$

13 is not co-prime of 26

So, we cannot find inverse of 13 mod 26

Therefore, it is not a valid key for affine cipher.

for 'input'

$$I = (8 \times 13 + 4) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$N = (13 \times 13 + 4) \bmod 26 = 173 \bmod 26 = 17 = R$$

$$P = (15 \times 13 + 4) \bmod 26 = 199 \bmod 26 = 17 = R$$

$$U = (20 \times 13 + 4) \bmod 26 = 264 \bmod 26 = 4 = E$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

ERROR

for 'alter'

$$A = (0 \times 13 + 4) \bmod 26 = 4 \bmod 26 = 4 = E$$

$$L = (11 \times 13 + 4) \bmod 26 = 147 \bmod 26 = 17 = R$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

$$E = (4 \times 13 + 4) \bmod 26 = 56 \bmod 26 = 4 = E$$

$$R = (017 \times 13 + 4) \bmod 26 = 225 \bmod 26 = 17 = R, \text{ ERROR}$$

Both the plaintext will produce the same ciphertext. The encryption is not one way and decryption is not possible as 13 is not a valid key.

7 Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$$

$$\det(K) = 3 \times 7 - 2 \times 5 = 21 - 10 = 11$$

11^{-1}

$$26 = 11(2) + 4 \quad 4 = 26 + 11(-2)$$

$$11 = 4(2) + 3 \quad 3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 - 3(1) = 1$$

$$26 + 4(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + (1(-1) + 4(2)) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(8) + 4(-7) = 1$$

$$26 - 7 = 19$$

Adjoint of the matrix

$$\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} = K^{-1} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the inverse of the following matrix whose entries are considered modulo 26: $K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

$$K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \pmod{26}$$

$$\det(K) = 11 \times 3 - 13 \times 2 = 33 - 26 = 7$$

$$7^{-1}$$

$$26 = 7(3) + 5 \quad 5 = 26 + 9(-3)$$

$$7 = 5(1) + 2 \quad 2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$26 + 7(-3) + 2(7 + 5(-1))(-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + (26 + 7(-3))(2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

Adjoint is

$$\text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26}, K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9) Is the key valid?

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

for a key to be valid - 1 to determinant should be coprime and should not be equal to 0

$$\det(k) = 2(21) - 14(13) = 42 - 182$$

$$= -140$$

$$-140 \bmod 26 = 16$$

$$\det k = 16 \bmod 26$$

16 is not coprime to 26

Therefore it will not have an inverse modulo 26

$\therefore$ The key $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$ is not valid

10) Encrypt using 2×2 Hill cipher for the plaintext "SOLVED" using

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$$

$$3 \times 1 - 7 \times 8 = -23 = 3, \therefore \text{it is a coprime}$$

S O L V E D

$$C = K \times P$$

$$P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26$$

$$166 \bmod 26 = 10 = K$$

$$280 \bmod 26 = 20 = U$$

$$\begin{matrix} L \\ V \end{matrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26}$$

$$201 \pmod{26} = 19 = T$$

$$308 \pmod{26} = 22 = W$$

$$\begin{matrix} E \\ D \end{matrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26}$$

$$36 \pmod{26} = 10 = K$$

$$61 \pmod{26} = 9 = J$$

SOLVED $\rightarrow$ K V T W K J